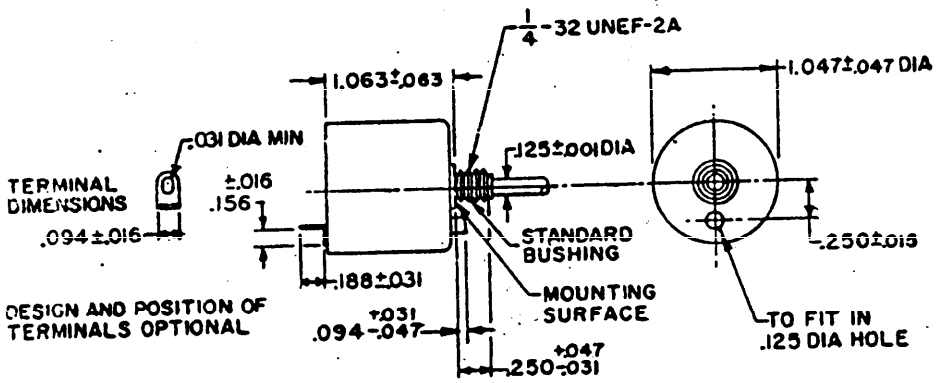
MIL-R-22/2C
14 February 1966
SUPERSEDING
MIL-R-22/2B
22 June 1965

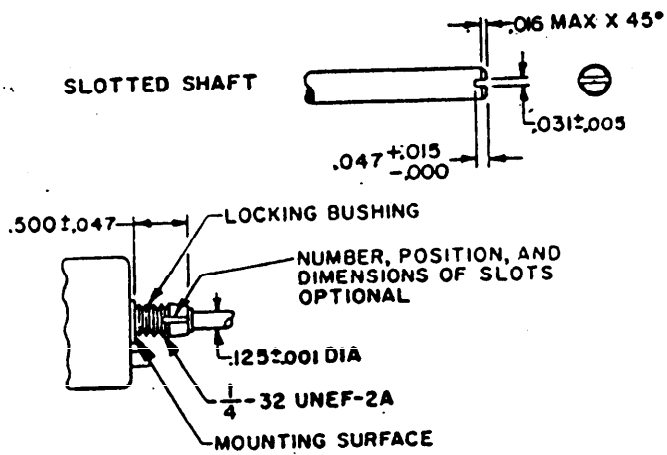
# MILITARY SPECIFICATION SHEET

## RESISTORS, VARIABLE (WIREWOUND, POWER TYPE)

## (ENCLOSED), STYLE RP07

The complete requirements for procuring the resistors described herein shall consist of this document and the latest issue of Specification MIL-R-22.




| INCHES | MM |
|---|---|
| .001 | .03 |
| .005 | .13 |
| .015 | .38 |
| .016 | .41 |
| .031 | .79 |
| .047 | 1.19 |
| .063 | 1.60 |
| .094 | 2.39 |
| .125 | 3.18 |
| .156 | 3.96 |
| .188 | 4.78 |
| .250 | 6.35 |
| .500 | 12.70 |
| 1.047 | 26.59 |
| 1.063 | 27.00 |

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

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FSC 5905

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TABLE I. — Style RP07.

| Type designation[1/] (for reference only) | Nominal total resistance value (ohms) | Maximum current (not to be exceeded on any portion of the winding) (amperes) | Part number[1/] |
|---|---|---|---|
| RP07---1R0KK | 1.0 | 2.50 | M22/02-0001--- |
| RP07---2R0KK | 2.0 | 1.77 | M22/02-0002--- |
| RP07---2R5KK | 2.5 | 1.58 | M22/02-0003--- |
| RP07---3R0KK | 3.0 | 1.44 | M22/02-0004--- |
| RP07---5R0KK | 5.0 | 1.12 | M22/02-0005--- |
| RP07---6R0KK | 6.0 | 1.02 | M22/02-0006--- |
| RP07---8R0KK | 8.0 | 0.88 | M22/02-0007--- |
| RP07---100KK | 10 | 0.79 | M22/02-0008--- |
| RP07---150KK | 15 | 0.65 | M22/02-0009--- |
| RP07---250KK | 25 | 0.50 | M22/02-0010--- |
| RP07---350KK | 35 | 0.42 | M22/02-0011--- |
| RP07---500KK | 50 | (C) 0.35 | M22/02-0012--- |
| RP07---750KK | 75 | 0.29 | M22/02-0013--- |
| RP07---101KK | 100 | 0.25 | M22/02-0014--- |
| RP07---151KK | 150 | 0.21 | M22/02-0015--- |
| RP07---201KK | 200 | 0.18 | M22/02-0016--- |
| RP07---251KK | 250 | 0.16 | M22/02-0017--- |
| RP07---351KK | 350 | 0.13 | M22/02-0018--- |
| RP07---501KK | 500 | 0.11 | M22/02-0019--- |
| RP07---751KK | 750 | 0.091 | M22/02-0020--- |
| RP07---102KK | 1,000 | 0.079 | M22/02-0021--- |
| RP07---152KK | 1,500 | 0.064 | M22/02-0022--- |
| RP07---252KK | 2,500 | 0.050 | M22/02-0023--- |
| RP07---352KK | 3,500 | 0.042 | M22/02-0024--- |

1/ The complete type designation and part number include symbols indicating electrical off position from table II, style of shaft and type of mounting from table III, and length of operating shaft from table IV (see example of part number).

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TABLE II. Electrical off position.

| Symbol | Electrical off position |
|---|---|
| 1 | No electrical off position. |
| 2 | Electrical off position at end of rotation of control knob in a counterclockwise direction. |
| 3 | Electrical off position at end of rotation of control knob in a clockwise direction. |

TABLE III. Style of shaft and type of mounting.

| Symbol | Style of shaft | 1/8-inch diameter shaft | |
|---|---|---|---|
| | | Standard bushing | Locking bushing |
| S | Slotted | X | --- |
| U | Slotted | --- | X |

TABLE IV. Length of operating shaft.

| Standard length of shaft measured from mounting surface of resistor. | | |
|---|---|---|
| Symbol | Slotted | |
| | 1/8-inch diameter shaft | |
| | Standard bushing | Locking bushing |
| | Inches, ±3/64 | Inches, ±3/64 |
| B | 5/8 | 5/8 |
| D | --- | 7/8 |
| H | 1-1/2 | |
| J | 2 | --- |
| K | 2-1/2 | --- |
| S | 3/8 | --- |

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REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Resistance value: See table I.

Resistance tolerance: ±10 percent.

Electrical off position: All three off positions applicable. See table II.

Shaft style and type of mounting: See table III.

Shaft length: See table IV.

Mechanical rotation: $300^{\circ}{}^{+5^{\circ}}_{-10^{\circ}}$.

Hardware: One 1/4-32 UNEF-2B corrosion-resistant, hexagonal mounting nut having a nominal thickness of 1/16 inch and measuring 5/16 inch across the hexagonal flats, and one corrosion-resistant, internal-tooth lockwasher having an outside nominal diameter of 0.402 inch, inside nominal diameter of 0.262 inch, and a nominal thickness of 0.013 inch shall be supplied. The locking nut for locking-bushing-type resistors shall be 5/32 inch thick, and shall measure 5/16 inch across the hexagonal flats; the thread size shall be 1/4-32 UNEF-2B.

Dielectric withstanding voltage:
At atmospheric pressure - Method 301 of MIL-STD-202. Test voltage 500 volts root mean square (rms).
At reduced pressure - Method 105 of MIL-STD-202. Test voltage - 250 volts.

DC resistance: Total and minimum. Method 303 of MIL-STD-202.

Total resistance within ±10 percent of the nominal total resistance specified in table I.

Minimum resistance shall not exceed 0.2 percent of the nominal total resistance value or 0.2 ohm, whichever is greater.

Torque:
Operating - Not less than 0.5 ounce-inches (oz-in), nor more than 6.0 ounce inches (oz-in). Stop - 5 lb-in. Locking - Shall withstand 20 oz-in without movement.

Low-temperature exposure:
Torque not in excess of 8 oz-in.

Power rating: 6.25 watts at 25°C.

Load life: Full wattage (see power rating, above) applied 1-1/2 hours on and 1/2 hour off for 1,000 hours. Change in total resistance not to exceed ±5 percent.

Vibration: Not applicable (see high frequency vibration).

Salt spray (corrosion): Method 101 of MIL-STD-202. Test condition letter B (48 hours).

Humidity (steady state): Method 103 of MIL-STD-202. Test condition letter B (96 hours). Change in total resistance shall not exceed ±10 percent.

Life (rotational): Method 206 of MIL-STD-202. Test condition letter A (500 cycles) for locking bushing types. Test condition letter C (5,000 cycles) for standard bushing types. Change in total resistance not to exceed ±5 percent.

Terminal strength: No mechanical damage with 20-pound load applied each terminal for 1 minute. Change in total resistance shall not exceed ±1 percent.

Shock: Method 202 of MIL-STD-202. Change in resistance shall not exceed ±10 percent due to contact arm movement and the total resistance change shall not exceed ±2 percent.

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Vibration, high frequency: Method 204 of MIL-STD-202. Test condition letter B (15G) 10 to 2,000 cps. Movement of contact arm not to result in a resistance change greater than ±10 percent. Total resistance change not to exceed ±2 percent. No mechanical damage.

Acceleration: 50G for 5 seconds. Change in resistance not to exceed ±10 percent due to contact arm movement, total resistance change not to exceed ±3 percent. No mechanical damage.

Part number: Consists of the basic number of this specification sheet, a dash number from table I, and electrical and mechanical characteristics taken from tables II, III, and IV.

EXAMPLE:

| M22/02- | 0001 | 1 | S | B |
|---|---|---|---|---|
| Basic document number (indicating MIL-R-22/2) | Dash number from table I (indicating 1.0 ohms) | Electrical off position from table II (indicating off position not used) | Style of shaft and type of mounting from table III (indicating standard bushing, slotted shaft) | Length of shaft from table IV (indicating 5/8-inch shaft) |

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH, WP
Air Force - 11, 17, 85

User activities:
Army - MO
Navy - MC

Preparing activity:
Navy - SH

(Project 5905-0570)

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